ZOO518P – APPLIED ENTOMOLOGY & PEST MANAGEMENT

Practical Exam Notes

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MCQs (Multiple Choice Questions)

1. Ecological importance of insects and their relative insect collection are critical for educating the public about: **BIODIVERSITY**.
2. Larval stages of butterflies or moths are usually **CATERPILLARS**.
3. **LIGHT TRAPS** are mainly used for attracting moths and other night flying insects.
4. Collected insects can be protected for longer time in insect collection box by putting **NAPHTHALENE** balls on the corners of the boxes.
5. A **COMPOUND MICROSCOPE** will be required for examining micro slides with magnification of 100 to 200 times or more.
6. Liquid preservation is done in **70%** ethyl alcohol + 4% normal saline.
7. Aphids, Whiteflies, Thrips prefer **YELLOW** color.
8. An appropriate binocular microscope should provide magnifications from **10 to 100** times.
9. Scanning Electron Microscope (SEM) which produce **THREE** dimensional images magnified by over 1000 times.
10. For morphological identification, binocular microscope should provide magnifications from **20 to 60** times.
11. **HAND PICKING** method is suitable for catching the large insects like beetles and grasshoppers.
12. Potassium Cyanide (KCN), Ethyl acetate, carbon tetrachloride and chloroform are commonly used for **KILLING** insects.
13. A single female lays and average of **200 to 500** eggs after fertilization and deposits inside the incubating chamber of the female cell.
14. **QUEEN EXCLUDER** is an optional piece of equipment and is used by less than 50% of beekeepers.
15. The **LARVA** is killed in the cocoon by steam or hot air at the chrysalis stage.
16. The silkworm caterpillar builds its **COCOON** by producing and surrounding itself with a long, continuous fiber or filament.
17. A weevil's **ROSTRUM** or elongated snout hosts chewing mouthparts instead of piercing mouthparts.
18. Lac insect is cultured on **KUSUM** trees.
19. Silk worm is cultured on **MULBERRY** trees.
20. Effective conservation and management of biodiversity depends in large part on our understanding of **TAXONOMY**.
21. The process of preserving insects for museum displays is called **SPECIMENIZATION**.
22. Most aerosols contain **LOW** percentage of active ingredients.

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23. Strategy is involved in using a combination of different pest control method of maximum effectiveness is called **INTEGRATED PEST MANAGEMENT (IPM).**
24. Scientific name of Babul is **ACACIA ARABICA**.
25. Scientific name of Ber (plum) is **ZIZYPHUS JUJUBE**.
26. The native species of honey bees include **APIS DORSATA**.
27. Palas is the common name of **BUTEA FRONDOSA**.
28. Common name of lac insects is **KERRIA LACCA**.
29. Which taxonomic rank represent the most specific level of classification? **SPECIES**
30. **MITES** have **EIGHT** legs as compared to the insects.
31. Pest control is managed by **INTEGRATED PEST MANAGEMENT**.

SHORT QUESTIONS

1. Which method is used to capture the active insects?

Active insects are captured using an aerial net, which consists of a light handle, a metal ring, and a fine mesh bag. The net is swept through the air to catch flying insects without damaging them. This method is widely used in entomological collections and insect surveys.

2. Write a note on sericulture and its importance in Pakistan industry.

Sericulture is the rearing of silkworms for the production of silk. The silkworm *Bombyx mori* is commonly used and feeds on mulberry leaves. The process involves rearing silkworms, collecting cocoons, and extracting silk fibers.

Importance in Pakistan Industry:

- Provides employment opportunities in rural areas.
- Generates income for farmers and small-scale industries.
- Supplies raw material to the textile and handicraft industries.
- Promotes cottage industries and rural development.
- Contributes to foreign exchange earnings through silk products.

3. Which plant tree is used in sericulture and lac culture?

Mulberry tree (*Morus* spp.) is used in sericulture, while Ber tree (*Ziziphus mauritiana*), Palas tree (*Butea monosperma*), and Kusum tree (*Schleichera oleosa*) are commonly used in lac culture.

4. Write a note on morphology of Aphids.

Aphids are small, soft-bodied insects belonging to the order Hemiptera. Their body is pear-shaped and divided into head, thorax, and abdomen. They possess long antennae, compound eyes, and piercing-sucking mouthparts used for feeding on plant sap. Aphids usually have two tube-like structures called cornicles on the dorsal side of the abdomen, which secrete defensive substances. They may be winged or wingless, and their color varies from green and yellow to black or brown. The legs are slender and adapted for crawling on plants.

5. Which is the best season in Pakistan for rearing silkworm?

Spring season (March–April) is the best season in Pakistan for rearing silkworms because mulberry leaves are abundant and environmental conditions are favorable for their growth and development.

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6. Differentiate between primary and secondary pest in stored grains.

Primary pests:

These pests attack and infest sound, whole grains directly. They can damage grains from inside as well as outside. Example: rice weevil, grain borer.

Secondary pests:

These pests attack already damaged, broken, or processed grains. They cannot infest whole grains directly and mainly feed on flour or broken parts. Example: red flour beetle, saw-toothed grain beetle.

7. What should we do before going on a field visit?

- ✓ Obtain permission from the concerned authority
- ✓ Understand the objective and route of the field visit
- ✓ Arrange necessary equipment (notebook, pen, camera, collection tools)
- ✓ Prepare a checklist for observations and data recording
- ✓ Ensure safety measures are followed
- ✓ Wear suitable clothing and footwear

8. Write scientific name of Silkworm.

Bombyx mori

9. What is the byproduct of Honeybee? What are its uses?

Bees produce several useful byproducts including beeswax, propolis, royal jelly, and bee venom in addition to honey.

Uses:

Beeswax: Used in candle making, cosmetics, ointments, polishes, and in pharmaceutical industries.
Propolis: Used in medicines and health products due to its antibacterial, antifungal, and healing properties.
Royal jelly: Used as a dietary supplement because it is rich in proteins, vitamins, and minerals and is believed to improve health and immunity.
Bee venom: Used in apitherapy for treating conditions like arthritis and joint pain.

10. Write uses of camera Lucida for taxonomy collection of insect.

- ✓ Camera lucida is used to obtain accurate drawings of insect specimens during taxonomic studies.
- ✓ It helps in producing true-to-scale illustrations of insect morphology.
- ✓ It is useful for drawing minute structural details like wings, antennae, mouthparts, and genitalia.
- ✓ It assists taxonomists in identification and classification of insects.
- ✓ It ensures high precision in scientific documentation of insect specimens in collections.

11. Write types of insect preservation methods to study their morphology.

Pinning method: Insects are mounted on entomological pins to preserve body structure.
Dry preservation method: Insects are dried and stored in insect boxes with naphthalene.
Wet preservation method: Insects are preserved in alcohol (70–90%) or formalin solution.
Slide mounting method: Small insects or body parts are mounted on glass slides using Canada balsam for microscopic study.
Freezing method: Insects are preserved at low temperatures for short-term storage.

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12.How cocoon is made from silkworm?

- ✓ The silkworm (*Bombyx mori*) feeds on mulberry leaves and grows through several larval stages.
- ✓ After full growth, it stops feeding and secretes a liquid protein from its silk glands.
- ✓ This liquid hardens on contact with air to form a fine silk thread.
- ✓ The larva moves its head in a figure-of-eight motion and continuously wraps the silk thread around its body.
- ✓ This process continues for 2–3 days until a complete protective covering called a cocoon is formed.

13.Why do you need apiary on your field?

- ✓ Apiary is needed in the field for pollination of crops, which increases yield and quality.
- ✓ It helps in improving fruit and seed production by enhancing cross-pollination.
- ✓ It supports beekeeping (apiculture) for honey production, providing income to farmers.
- ✓ It promotes biodiversity and ecological balance in the agricultural ecosystem.
- ✓ It increases overall agricultural productivity and profitability of the field.

14.Why pest should be controlled?

- ✓ Pests should be controlled to prevent damage to crops and stored grains.
- ✓ They reduce yield and quality of agricultural products.
- ✓ Pest control helps in preventing spread of plant diseases.
- ✓ It ensures food security and reduces economic losses.
- ✓ It improves overall agricultural productivity and profitability.

15.Write identification and control of Whitefly of cotton.

Identification:

Whitefly (*Bemisia tabaci*) is a small, soft-bodied, white insect usually found on the underside of cotton leaves. Both nymphs and adults suck plant sap, causing yellowing, leaf curling, stunted growth, and reduced yield. They also excrete sticky honeydew, which leads to the growth of black sooty mold on leaves and reduces photosynthesis.

Control:

Whitefly can be controlled through integrated pest management (IPM). Cultural methods include removal of infested plant parts and weeds. Physical control involves using yellow sticky traps. Biological control includes encouraging natural enemies such as ladybird beetles and parasitoid wasps. Chemical control is done by spraying recommended insecticides like imidacloprid or acetamiprid when infestation exceeds economic threshold levels.

16.Why is there a need of pesticides?

- ✓ Pesticides are needed to control harmful pests, insects, weeds, and disease-causing organisms.
- ✓ They help protect crops from damage and reduce yield losses.
- ✓ Pesticides improve the quality and quantity of agricultural produce.
- ✓ They help in preventing the spread of plant diseases.
- ✓ They protect stored grains and food products from pest infestation.
- ✓ Their use contributes to higher agricultural productivity and food security.

17.Why insects are well arranged and displayed in insect museums?

- ✓ Insects are well arranged and displayed in insect museums for easy identification and classification.
- ✓ Proper arrangement helps in the study of insect morphology, diversity, and taxonomy.
- ✓ It preserves specimens for future scientific research and reference.
- ✓ It facilitates teaching and learning for students and researchers.
- ✓ Well-organized displays help in educating the public about insect biodiversity and their importance.

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18. Why insects are categorized in museums?

- ✓ Insects are categorized in museums to facilitate easy identification and classification.
- ✓ Categorization helps in the systematic study of insect diversity and taxonomy.
- ✓ It allows researchers to compare related species and understand their relationships.
- ✓ It improves the organization and management of insect collections.
- ✓ It serves as a valuable resource for research, education, and biodiversity conservation.

19. Sweep net method is used for which type of insect collection?

The sweep net method is used for collecting insects that live on grasses, crops, shrubs, and other low vegetation. It is particularly effective for capturing small and active insects such as aphids, leafhoppers, bugs, beetles, and small flies. The net is swept through the vegetation to collect insects for study and identification.

20. Write some uses of lac.

Lac is a natural resin secreted by lac insects and has many commercial uses. It is used in the manufacture of varnishes, polishes, paints, and printing inks. Lac is also used for making bangles, toys, decorative items, sealing wax, and electrical insulating materials. In the pharmaceutical and food industries, it is used as a coating material for tablets and confectionery products.

21. Write importance of museums for insect study.

Importance of Museums for Insect Study

- Preserve insect specimens for long-term scientific study.
- Help in the identification and classification of insects.
- Provide reference collections for taxonomic research.
- Record insect biodiversity of different regions and time periods.
- Support ecological and environmental studies.
- Help detect changes in insect populations and distributions.
- Serve as educational resources for students and researchers.
- Facilitate the discovery and description of new insect species.
- Promote public awareness about the importance of insects.
- Preserve rare, endangered, and extinct insect specimens for future generations.

LONG QUESTIONS

1. How the insects are positioned under the microscope?

Insects are carefully placed on the stage of a stereoscopic microscope using forceps, brushes, or insect pins. The specimen is positioned so that important structures such as the head, antennae, wings, legs, and abdomen are clearly visible. The insect may be viewed from different angles by gently adjusting its position. Proper lighting and focus are maintained to observe morphological features accurately for identification and study.

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2. Write some methods of insect prevention.

Insect prevention refers to the measures taken to reduce or stop insect infestation and the damage caused by insects. The main methods of insect prevention are:

1. Sanitation Method

Maintaining cleanliness in homes, farms, and storage areas helps prevent insect breeding. Proper disposal of garbage, removal of stagnant water, and regular cleaning reduce insect populations.

2. Physical Method

Physical barriers such as mosquito nets, window screens, and insect traps are used to prevent insects from entering houses or damaging crops. Hand-picking of insects is also a physical control method.

3. Biological Method

Natural enemies of insects, such as predators, parasites, and pathogens, are used to control harmful insect populations. This method is environmentally friendly and reduces the need for chemicals.

4. Chemical Method

Chemical insecticides and pesticides are applied to kill or control insects. These chemicals should be used carefully and according to recommended doses to avoid environmental pollution.

5. Cultural Method

Agricultural practices such as crop rotation, timely sowing, proper irrigation, and removal of infected plants help prevent insect attacks and reduce their spread.

3. Discuss locust attack in south Punjab.

Introduction

A locust attack is the sudden invasion of large swarms of locusts that feed on crops and vegetation. South Punjab has experienced several locust outbreaks, causing serious damage to agriculture and the economy.

Effects of Locust Attack

Locust swarms rapidly consume crops, vegetables, fruit plants, and fodder. Major crops affected in South Punjab include wheat, cotton, maize, and sugarcane. These attacks lead to reduced crop yields, economic losses for farmers, and food security concerns.

Causes of Outbreaks

Locust populations increase rapidly under favorable environmental conditions such as heavy rainfall, abundant vegetation, and suitable breeding grounds. Swarms often migrate from neighboring regions and spread across agricultural areas.

Control Measures

Locust control involves monitoring breeding areas, spraying insecticides, using ground and aerial control operations, and coordinating efforts among agricultural departments and farmers. Early detection and rapid response are essential to prevent severe damage.

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4. Write some methods of insect collection.

Introduction

Insect collection is an important part of entomological studies. Different methods are used to collect insects according to their habitat, behavior, and mode of life.

1. Hand Collection: Insects are collected directly by hand, forceps, or brushes. This method is suitable for large and slow-moving insects.

2. Sweep Net Method: A sweep net is used to collect insects from grasses, shrubs, and crop fields. The net is swept through vegetation to capture insects living on plants.

3. Aerial Net Method: An aerial net is used to catch flying insects such as butterflies, dragonflies, and bees. The collector swings the net through the air to capture the insects.

4. Light Trap Method: A light source is used at night to attract insects, especially moths and other nocturnal species. The insects are then collected from around the light.

5. Pitfall Trap Method: A container is buried in the ground so that crawling insects fall into it. This method is commonly used for beetles, ants, and other ground-dwelling insects.

6. Aspirator Method: An aspirator is a device used to collect small insects by sucking them into a collection tube without causing damage.

5. How cocoon is produced by larva instar of Bombyx mori?

Introduction

Bombyx mori (silkworm) is an economically important insect that produces silk. The cocoon is formed during the last larval instar stage before pupation.

Cocoon Formation

After completing its feeding period, the mature fifth-instar larva stops eating and begins searching for a suitable place to spin a cocoon. The larva possesses a pair of silk glands that secrete a protein called fibroin. This fibroin is coated with another protein called sericin, which acts as a natural gum.

The larva moves its head in a figure-eight pattern and continuously releases a silk filament from its spinneret. The silk thread hardens when exposed to air and gradually forms a protective covering around the larva. The spinning process usually takes about 2–3 days to complete.

Importance of Cocoon

The cocoon protects the developing pupa from environmental conditions and predators. It is also the primary source of natural silk used in the textile industry.

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6. Discuss different types of habitats for insect collection.

Insects are found in almost all environments. Different habitats support different types of insects, so understanding habitats is important for effective insect collection.

1. Terrestrial Habitats

These include land-based environments such as fields, forests, gardens, and grasslands. Many insects like beetles, ants, butterflies, and grasshoppers are commonly found here.

2. Aquatic Habitats

These include freshwater and sometimes marine environments such as ponds, lakes, rivers, and wetlands. Insects like dragonfly nymphs, mosquito larvae, and water beetles live in water.

3. Soil and Subterranean Habitats

These insects live in or under the soil. Examples include ants, termites, and many larvae of beetles. They are usually collected using pitfall traps or soil sampling methods.

4. Aerial Habitats

These insects spend most of their life flying in the air. Examples include flies, bees, butterflies, and dragonflies. They are collected using aerial nets and light traps.

5. Parasitic Habitats

Some insects live on or inside other organisms as parasites. Examples include lice, fleas, and some mites. They are collected from host animals or humans.

7. Write different positions of insects for microscopic examination.

For proper study and identification, insects are examined under a stereoscopic or compound microscope in different positions. Correct positioning helps in observing all important morphological features clearly.

1. Dorsal Position: In this position, the insect is placed with its back (dorsal side) facing upward. It is used to observe structures like wings, body segmentation, and dorsal patterns.

2. Ventral Position: The insect is placed with its underside (ventral side) facing upward. This position helps in studying mouthparts, legs, and abdominal structures.

3. Lateral Position: The insect is kept on its side. It is useful for observing body shape, wing attachment, and side structures of the head and thorax.

4. Frontal Position: In this position, the head is focused from the front. It is mainly used to study antennae, eyes, and mouthparts in detail.

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8. How is lac produced by lac insect?

Introduction

Lac is a natural resin produced by the lac insect (Kerria lacca). It is commercially important and used in varnishes, dyes, cosmetics, and polish materials.

Feeding and Secretion

The female lac insect feeds on the sap of host plants such as palas, ber, and kusum. While feeding, it sucks plant juice and converts it into a resinous secretion inside its body.

Excretion of Resin

The insect excretes this processed material through special glands present in its body. This secretion comes out as a sticky fluid that spreads over the body surface.

Formation of Lac Covering

On exposure to air, the sticky secretion hardens and forms a protective shell around the insect and the twig of the host plant. This hardened layer is called "lac."

9. Write advantages of wettable formulation of pesticides.

Wettable powder formulation is a type of pesticide that is mixed with water before application. It is commonly used in agriculture for effective pest control.

1. Easy to Apply

Wettable formulations can be easily mixed with water and sprayed using simple spraying equipment.

2. Good Coverage

When sprayed, it spreads evenly on plant surfaces, providing better coverage of pests and leaves.

3. Effective Control

It gives good insect control because the fine particles stick to plant surfaces after drying.

4. Economical

Wettable powders are generally cheaper compared to liquid formulations and are cost-effective for farmers.

5. Longer Shelf Life

They are more stable during storage and have a longer shelf life compared to many liquid pesticides.

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10.Write functions of Queen Bee excluder.

A queen bee excluder is a specially designed grid or mesh used in beekeeping to control the movement of the queen bee within the hive.

1. Restricts Queen Movement: It allows worker bees to pass through but prevents the queen bee from entering the honey supers.

2. Separates Brood and Honey Areas: It keeps the queen in the brood chamber, ensuring that eggs are laid only in the lower part of the hive, while honey is stored in upper boxes.

3. Improves Honey Quality: By preventing egg-laying in honey supers, it ensures that honey remains clean and free from brood contamination.

4. Facilitates Honey Harvesting: It makes it easier for beekeepers to collect honey without disturbing the queen or brood area.

5. Hive Management: It helps in better management of the colony by controlling the distribution of bees within different sections of the hive.

11.How insects are different from arachnids?

Insects and arachnids are both arthropods, but they differ in body structure, number of legs, and other features.

1. Body Division: Insects have three body parts: head, thorax, and abdomen. Arachnids have two body parts: cephalothorax and abdomen.

2. Number of Legs: Insects have three pairs of legs (6 legs). Arachnids have four pairs of legs (8 legs).

3. Antennae: Insects possess one pair of antennae. Arachnids do not have antennae.

4. Wings: Most insects have wings (one or two pairs), while arachnids never have wings.

5. Eyes: Insects usually have compound eyes. Arachnids have simple eyes.

6. Examples: Insects include butterflies, bees, and ants. Arachnids include spiders, scorpions, and ticks.

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